RAG TRAINING DOCUMENT

Climate-Smart Agriculture Research

Smallholder Potato Farmers in Kenya

Gilgil Sub-County, Nakuru County, Kenya | 2024

Handbook of Nature-Based Solutions to Mitigation and Adaptation to Climate Change

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This document is structured for use in a Retrieval-Augmented Generation (RAG) application. All sections are self-contained, clearly labelled, and optimised for accurate retrieval and contextual injection. Key findings, statistics, and definitions are stated as discrete, queryable facts. Cross-topic relationships are explicitly noted to support multi-step reasoning.

1. Paper Identity & Metadata

1.1 Full Citation

Waaswa, A., Aniche, A.-A., Nkurumwa, A. O., Kibe, A. M., & Kipkemoi, N. J. (2024). *Exploring the Future of Agriculture Under Climate Change: The Potentials of Climate-Smart Agriculture Among Smallholder Farmers in Kenya.* In W. Leal Filho et al. (Eds.), *Handbook of Nature-Based Solutions to Mitigation and Adaptation to Climate Change*. Springer. https://doi.org/10.1007/978-3-030-98067-2_146-1

1.2 Authors and Affiliations

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1.3 Publication Context

- Published in 2024 in the *Handbook of Nature-Based Solutions to Mitigation and Adaptation to Climate Change* (Springer).
- Funded by the MasterCard Foundation through RUFORUM (Regional Universities Forum for Capacity Building in Agriculture).
- Ethics approval: Kenya's NACOSTI, licence no. NACOSTI/P/21/9627.
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2. Study Context and Background

2.1 Research Problem

Climate variability — including low precipitation, flooding, high temperatures, prolonged sunshine, and delayed rainfall — threatens agricultural productivity, leading to food insecurity and chronic poverty, especially among resource-constrained smallholder farmers. Rain-fed agriculture, which dominates small-scale farming in Africa, is particularly susceptible. The study tests the hypothesis that adopting Climate-Smart Agriculture (CSA) is a significant strategy for climate change adaptation.

2.2 Study Area: Gilgil Sub-County

Attribute	Details
Location	Gilgil Sub-County, Nakuru County, Kenya

Attribute	Details
Coordinates	36° 10′ 00″ E, 0° 40′ 00″ S to 35° 30′ 00″ E, 1° 00′ 00″ S
Agro-Ecological Zone	Zone III of Kenya
Area	1,348.43 square kilometres
Total Population	171,839
Annual Rainfall	500 – 870 mm
Key Crops	Maize, beans, and potatoes (cover 86.4% of arable land)
Target Population	15,359 smallholder potato farmers (2019 agricultural census)
Accessible Population	10,889 potato farmers (Morendat ward: 4,287; Mbaruk/Eburu ward: 6,602)
Selection Rationale	High susceptibility to climate change effects; active CSA interventions by SNV, CaWSA-C, CARP+

2.3 Kenyan CSA Policy Context

- Kenya is vulnerable to climate change and has implemented multiple national frameworks.
- National Climate Change Response Strategy (NCCRS) launched in 2010.
- National Climate Change Action Plan (NCCAP 2013–2017) launched in 2013.
- Kenya Climate-Smart Agriculture Strategy (KCSAS) launched in 2017; covers 2017–2026.
- FAO concept of CSA first used in 2009; followed by Wageningen Statement, African Agricultural Ministers' call to action, and COP 17 in Durban 2011.
- FAO launched EPIC (Economics and Policy Innovations for Climate-Smart Agriculture) programme in 2013.
- Global Alliance for Climate-Smart Agriculture (GACSA) launched at UN Climate Summit in New York in 2013.
- Active programmes in study area: Climate and Water Smart Agriculture Centre (CaWSA-C), Community Action Research Project (CARP+).

3. Climate-Smart Agriculture: Definition and Practices

3.1 Definition of CSA

CSA is defined as methods, management strategies, and technical developments that sustainably increase productivity, improve resilience, lower greenhouse gas emissions, and better achieve food security and development objectives (Chandra et al. 2018). It is based on a 'triple-win' impact: increasing production, improving resilience, and lowering emissions. Farm-level CSA techniques are described as deliberate agricultural adaptation or environmental management methods applied on plots owned, leased, or used by a household.

3.2 CSA Practices Studied

This study focused on 14 specific CSA practices as defined by Waaswa et al. (2022):

#	CSA Practice	Category
1	Drought-tolerant and high-yielding crop varieties (improved varieties)	Varieties
2	Synthetic fertilizers	Inputs
3	Rainwater harvesting and storage	Water management
4	Agroforestry	Land management
5	Irrigation	Water management
6	Mulching	Soil management
7	Composting	Soil management
8	Terracing	Soil conservation
9	Potato apical rooted cuttings (potato seedlings)	Seed technology
10	Potato mini-tubers	Seed technology
11	Crop rotation	Agronomy
12	Intercropping	Agronomy
13	Drainage management	Water management
14	Minimum tillage / furrow-ridge planting	Soil management

4. Methodology

4.1 Research Design

- Cross-sectional survey design.
- Data collected using a structured researcher-administered questionnaire.
- Analysis: descriptive statistics (frequencies and percentages) and binary logistic regression.
- Software: SPSS version 28.
- Significance level: 5% ($p < 0.05$).

4.2 Sampling

Parameter	Value
Sampling formula	Nassiuma (2000): $n = NC^2 / (C^2 + (N-1)e^2)$
Coefficient of variation (C)	21%
Standard error (e)	0.02
Calculated sample size (n)	109
Adjusted sample size (+ 10% non-response)	120

Parameter	Value
Wards sampled	Mbaruk/Eburu (n=73) and Morendat (n=47)
Sampling method	Proportionate random sampling by ward; simple random sampling for respondents
Questionnaire reliability (Cronbach Alpha)	0.805 (threshold: >0.70)
Pilot study location	Mauche Ward, Njoro Sub-County, Nakuru County (30 farmers)

4.3 Hypothesis Tested

Null hypothesis (H0): Adopting CSA is not a significant strategy for climate change adaptation. **Alternative hypothesis (H1):** Adopting CSA is a significant strategy for climate change adaptation. Binary logistic regression was used to test this at 5% significance level.

5. CSA Adoption Rates

5.1 Adoption Levels by Practice

Results show that 64.56% of potato farmers on average adopted CSA practices. Adoption rates by practice:

CSA Practice	Adoption Rate (%)
Synthetic fertilizers	95.0 (highest adopted)
Intercropping	89.2
Agroforestry	85.0
Rainwater harvesting and storage	83.3
Crop rotation	83.3
Terracing	75.0
Composting	75.8
Furrow/ridge planting	74.2
Minimal tillage	72.5
Drainage management	70.8
Mulching	64.2
Improved crop varieties	59.2
Irrigation	31.7
Apical rooted cuttings (potato seedlings)	7.5
Mini-tubers	1.7 (least adopted)

5.2 Explanation of Adoption Patterns

High adoption – Synthetic fertilizers (95%)

- Release nutrients quickly and produce high yields to meet food demand.
- Can be applied instantly; organic fertilizers require time to decompose and are bulky.

High adoption – Crop rotation, agroforestry, intercropping, rainwater harvesting (>80%)

- Ensure differential nutrient uptake; improve soil fertility; reduce chemical fertilizer needs.
- Agroforestry improves land productivity, provides hospitable microclimate, carbon content.
- Water harvesting increases yields (Parrott & Marsden 2002 cited Senegal millet/peanut example).

Low adoption – Potato mini-tubers (1.7%) and apical rooted cuttings (7.5%)

- Lack of technical know-how among farmers.
- High cost of purchasing seedlings.
- Inability to reinvent (replicate) these practices independently.
- Require additional investment and higher recurring costs compared to other CSA practices.
- Management gap and technological gap are the two contributors to the productivity disparity.

6. Potentials of CSA Practices (Farmer-Reported)

Of the 64.56% of farmers who adopted CSA, the following proportions reported adaptation benefits:

Adaptation Potential	Farmers Reporting 'Yes'	Frequency (n)	% of adopters
Increasing incomes	Most reported	114	95.0
Improving soil fertility		113	94.2
Increasing yields		113	94.2
Reducing labour requirements		111	92.5
Increasing product quality		107	89.2
Reducing soil erosion		106	88.3
Reducing input expenses		105	87.5
Reducing pests and disease infestation		104	86.7
Increasing livelihood diversification		101	84.2
Watering animals		95	79.2
Ensuring production all year round	Least reported	68	56.7

7. Statistical Findings – Binary Logistic Regression

7.1 Hypothesis Test Result

Binary regression analysis revealed that CSA is a statistically significant climate change adaptation strategy: **Wald χ^2 = 49.417, df = 1, p < 0.001.** Implementing CSA increased farmers' chances to adapt to climate change by **19 times** compared to non-adopters.

7.2 Regression Coefficients – All Potentials

Potential	B	SE	Wald stat	df	p-value	Exp(B) = Odds Ratio
Improving soil fertility	0.871	0.795	1.200	1	0.273	2.389
Increasing yields	0.194	0.865	0.050	1	0.822	1.214
Increasing incomes	0.429	0.893	0.231	1	0.631	1.536
Reducing soil erosion	1.986	0.609	10.645	1	0.001 *	7.286
Ensuring production all year round	0.720	0.427	2.843	1	0.092	2.055
Increasing livelihood diversification	2.371	0.561	17.882	1	<0.001 *	10.706
Reducing labour requirements	0.442	0.741	0.356	1	0.551	1.556
Reducing input expenses	1.792	0.581	9.518	1	0.002 *	6.000
Reducing pests and disease infestation	0.693	0.566	1.498	1	0.221	2.000
Increasing product quality	1.822	0.618	8.682	1	0.003 *	6.182
Watering animals	0.506	0.497	1.039	1	0.308	1.659

* Statistically significant at p < 0.05.

7.3 Four Statistically Significant Potentials

The binary regression identified four CSA potentials that significantly contribute to climate change adaptation:

Potential	Odds Ratio (Exp B)	Wald	p-value	Interpretation
Reducing soil erosion	7.286	10.645	0.001	CSA adopters are 7.3x more likely to reduce soil erosion
Increasing livelihood diversification	10.706	17.882	<0.001	CSA adopters are 10.7x more likely to diversify livelihoods
Reducing input expenses	6.000	9.518	0.002	CSA adopters are 6.0x more likely to reduce input expenses
Increasing product quality	6.182	8.682	0.003	CSA adopters are 6.2x more likely to improve product quality

8. Detailed Analysis: Four Significant CSA Potentials

8.1 Reducing Soil Erosion (Odds Ratio = 7.286, p = 0.001)

CSA reduces soil erosion among adopters 7.286 times more than non-adopters.

Mechanisms:

- Low/no-till farming leaves agricultural residues on the field after harvest, acting as protective cover.
- Cover crops planted during fallow periods stabilise soil with roots and add organic matter.
- Agroforestry — trees and shrubs create natural windbreaks, slowing wind and preventing erosion.
- Terracing and contour ploughing on sloping land minimise erosive power of water runoff.
- Soil health practices (organic matter addition, fertiliser management, reducing soil compaction) minimise erosion.
- USDA (2021): perennial cover crops in strip systems increase soil carbon sequestration, build soil health, reduce erosion.

Caution:

- Reforestation/afforestation in sensitive ecosystems or steep slopes may paradoxically worsen soil erosion if implemented incautiously.

8.2 Increasing Livelihood Diversification (Odds Ratio = 10.706, p < 0.001)

CSA increases livelihood diversification among adopters 10.706 times more than non-adopters. This is the strongest effect among the four significant potentials.

Mechanisms:

- Crop diversification improves farmer and landscape resilience.
- Commercialisation and enhanced farm profitability enable smallholders to expand and scale.
- Surplus income from CSA-driven higher yields can fund off-farm businesses (agri-processing, service-based ventures).
- Agroforestry and sustainable forest management provide timber, fuelwood, fruits, nuts, and forest products.
- Eco-tourism and nature-based business opportunities created through environmentally sound CSA behaviours.
- Community-based projects (cooperatives, community gardens, market linkages) open shared livelihood prospects.

Limitation:

- Non-agricultural livelihood transformation may merely be a poor coping strategy if smallholders lack access to financial resources, land, water, and technology required to diversify.

8.3 Reducing Input Expenses (Odds Ratio = 6.000, p = 0.002)

CSA reduces input expenses among adopters 6.0 times more than non-adopters.

Mechanisms:

- Composting and crop residues as mulch or green manure reduce demand for chemical fertilisers.
- Integrated Pest Management (IPM) reduces reliance on chemical pesticides.
- Contour ploughing and agroforestry prevent soil erosion and water loss, reducing recovery costs after drought or flood.
- CSA enables access to premium markets that reward sustainable practices (certifications, labelling), helping cover input costs.
- Mulching and no-till farming reduce labour-intensive weeding and ploughing, reducing labour costs.

- Khatri-Chhetri et al. (2016) found smallholder farmers significantly benefited financially from CSA.

Caveat:

- Early adopters face yield risk, equipment costs, and input costs as initial barriers.
- Monast and Yeoman (2023) note early adopters may be shut out of ecosystem markets due to extra requirements.

8.4 Increasing Product Quality (Odds Ratio = 6.182, p = 0.003)

CSA increases product quality among adopters 6.182 times more than non-adopters.

Mechanisms:

- Organic management practices improve biotic elements of soil health (microbial activity and diversity), improving crop quality.
- Balanced nutrient management (soil testing, precision inputs) ensures crops receive adequate nutrients — improves nutritional value.
- Rainwater harvesting and drip irrigation ensure adequate water supply without waterlogging — tastier, more flavorful produce.
- Appropriate post-harvest handling (storage, transportation) maintains produce in better shape.
- Climate-smart certifications and traceability systems boost consumer confidence and product perception.
- 'Climate-smart' labelled products may command higher prices as consumer awareness of sustainability grows.
- Reilly et al. (2013): organic fertiliser in onion fields resulted in soil-suppressive functional groups dominating over pathogens.
- Reganold et al. (2010): CSA measures on strawberry farms (organic amendments) resulted in higher-quality fruit.

Limitation:

- Extreme weather events (prolonged droughts, floods) can still affect crop quality despite CSA adoption.
- Limited access to inputs (quality seeds, fertilisers, equipment) can hinder product quality improvement in some regions.

9. Climate Change Effects on Agriculture and CSA Responses

The study's literature review provides a comprehensive matched framework of climate change effects on agricultural production and corresponding CSA responses. Key matched pairs:

Climate Change Effect	Key CSA Response
Declining soil fertility (faster organic matter decomposition, nutrient leaching)	No-till farming, organic amendments (manure, compost), agroforestry, crop rotation, terracing, precision fertilis
Diminished yields per unit area (heat stress, drought, waterlogging, pest/disease shifts, reduced	Soil health practices, crop diversification, stress-tolerant varieties, climate-resilient varieties, IPM, precision farm
Soil erosion (extreme weather, reduced vegetation cover, droughts, grazing)	Cover crops, terracing, afforestation/reforestation, low-till farming, cover crops, agroforestry wind
Inability to produce year-round (shifting growing seasons, water scarcity, soil degradation)	Crop diversification, improved irrigation, agroforestry, hardy varieties, greenhouses/polytu
Reduced livelihood options (lower farm income, unemployment)	Diversified income via CSA, off-farm agroforestry products, eco-tourism, value addition, c

Climate Change Effect	Key CSA Response
Increased labour requirements (more pest/weed pressure, erratic rainfall)	Mulching, crop diversification, irrigation, soil conservation, farm management software/apps
Increased input expenses (irrigation infrastructure, pesticides, fertilisers)	Combining synthetic inputs with agroforestry (soil conservation), premium market access via sustainabi
Increased pests and disease infestation (longer dry seasons favouring pests)	IPM (crop rotation, biological control, judicious pesticide use), biopesticides/biocontrol ager
Reduced product quality (heat stress, water stress, nutrient imbalances)	Balanced nutrient management, IPM, appropriate irrigation, post-harvest handling, certific
Reduced water for agriculture (droughts, altered rainfall patterns)	Drought-tolerant varieties, drip/sprinkler/subsurface irrigation, rainwater harvesting, water

10. Conclusions and Recommendations

10.1 Key Conclusions

- Average of **64.56%** of smallholder potato farmers in Gilgil Sub-County adopted CSA practices.
- Synthetic fertilisers were the most adopted CSA (95%); mini-tubers were the least adopted (1.7%).
- Binary regression confirmed: CSA is a statistically significant climate change adaptation strategy (Wald chi-squared = 49.417, df = 1, $p < 0.001$).
- Implementing CSA increases farmers' chances of adapting to climate change by **19 times** compared to non-adopters.
- CSA significantly contributes to adaptation through four pathways: reducing soil erosion, increasing livelihood diversification, reducing input expenses, and increasing product quality.
- These four benefits create a ripple of positive impacts: preserved soil fertility, spread income risk, sustainable farming motivation, and reduced waste along the supply chain.

10.2 Recommendations

Recommendation	Rationale
Use experiential learning approaches	Allows farmers to actively engage with new CSA practices and technologies, leading to better unde
Establish linkages between research institutions and farmers	Facilitates exchange of information/expertise; creates dynamic feedback loop that fosters CSA acce
Capitalise on progressive farmers as role models	Progressive farmers can share experiences, demonstrate CSA benefits, and provide practical guida
Address barriers to potato mini-tubers adoption	Technical training, cost support, and reinvention capacity needed to close the technological and
Promote CSA through government extension, CAHWs, CAPEs, and SNV, CanWaCC, partnerships	Leverages existing networks and government extension officers model effective delivery channels

11. Glossary of Key Terms

Term	Definition
CSA – Climate-Smart Agriculture	Methods, management strategies, and technical developments that sustainably increase productivit
Triple-win	The three core CSA objectives: (1) increase production/food security, (2) improve resilience/adapta

Term	Definition
KCSAS	Kenya Climate-Smart Agriculture Strategy (2017–2026). Guides changes to agricultural systems thr
NCCAP	National Climate Change Action Plan (Kenya, 2013–2017). Part of Kenya's climate governance fram
NCCRS	National Climate Change Response Strategy (Kenya, 2010).
CaWSA-C	Climate and Water Smart Agriculture Centre. Active in Gilgil Sub-County.
CARP+	Community Action Research Project. Active in Gilgil Sub-County.
RUFORUM	Regional Universities Forum for Capacity Building in Agriculture. Provided funding through MasterC
NACOSTI	National Commission for Science, Technology, and Innovation (Kenya). Issued ethics approval.
Binary logistic regression	Statistical method used to predict a binary outcome (adapts/does not adapt) from predictor variable
Odds ratio / Exp(B)	How many times more likely CSA adopters are to experience a given outcome compared to non-ad
Wald statistic	Test statistic in logistic regression used to assess whether a predictor is statistically significant.
IPM – Integrated Pest Management	Approach combining biological control, crop rotation, prudent pesticide use to manage pests with m
Apical rooted cuttings	Potato seedlings produced through tissue culture; disease-free seed material. Classified as a CSA p
Mini-tubers	Small potato seed tubers produced through modern seed technology. Classified as a CSA practice.
Agroforestry	Integration of trees and shrubs into agricultural landscapes; improves soil fertility, provides windbrea
Smallholder farmer	Farmer operating on a small area of land, often with limited capital, relying on family labour and faci
Agro-ecological zone	Classification of land areas by climate, soil, and vegetation to guide agricultural recommendations.
Cronbach Alpha	Measure of questionnaire reliability. Score of 0.805 obtained (threshold for reliability: >0.70).
Coefficient of variation (C)	Statistical measure of relative variability used in Nassiuma sample size formula. Set at 21% in this s

12. Quick Reference Facts for RAG Retrieval

Discrete, queryable factual statements for high-precision retrieval:

- 64.56% of smallholder potato farmers in Gilgil Sub-County adopted CSA practices on average.
- Synthetic fertilisers were the most adopted CSA practice at 95%.
- Potato mini-tubers were the least adopted CSA practice at 1.7%.
- Apical rooted cuttings (potato seedlings) had a 7.5% adoption rate — second lowest.
- CSA adoption increases farmers' chances of adapting to climate change by 19 times (Wald chi-squared = 49.417, df = 1, $p < 0.001$).
- Four CSA potentials are statistically significant climate change adaptation pathways: soil erosion reduction, livelihood diversification, input expense reduction, and product quality improvement.
- CSA reduces soil erosion 7.286 times more among adopters (Wald = 10.645, $p = 0.001$).

- CSA increases livelihood diversification 10.706 times more among adopters (Wald = 17.882, $p < 0.001$). This is the strongest effect.
- CSA reduces input expenses 6.0 times more among adopters (Wald = 9.518, $p = 0.002$).
- CSA increases product quality 6.182 times more among adopters (Wald = 8.682, $p = 0.003$).
- 95% of CSA adopters reported CSA helped increase income.
- 94.2% of CSA adopters reported CSA helped improve soil fertility and increase yields.
- 92.5% of CSA adopters reported CSA reduced labour requirements.
- 89.2% of CSA adopters reported CSA increased product quality.
- 88.3% of CSA adopters reported CSA reduced soil erosion.
- 87.5% of CSA adopters reported CSA reduced input expenses.
- 86.7% of CSA adopters reported CSA reduced pests and disease infestation.
- 84.2% of CSA adopters reported CSA increased livelihood diversification.
- 79.2% of CSA adopters reported CSA helped with watering animals.
- 56.7% of CSA adopters reported CSA ensured production all year round — the least reported benefit.
- Study location: Gilgil Sub-County, Nakuru County, Kenya. Agro-ecological Zone III. Annual rainfall 500–870 mm.
- Target population: 15,359 smallholder potato farmers in Gilgil Sub-County (2019 census).
- Sample size: 120 farmers (73 from Mbaruk/Eburu ward, 47 from Morendat ward).
- Questionnaire reliability: Cronbach Alpha = 0.805.
- Kenya launched KCSAS (Climate-Smart Agriculture Strategy) in 2017, covering 2017–2026.
- FAO first used the concept of CSA in 2009. GACSA launched at the 2013 UN Climate Summit in New York.
- Low adoption of potato mini-tubers and apical cuttings is attributed to lack of technical know-how, high seedling costs, and inability to reinvent practices.
- Rainwater harvesting adopted by 83.3% of farmers; agroforestry by 85%; intercropping by 89.2%.
- Rain-fed agriculture dominates small-scale farming in Africa and is particularly susceptible to climate change.
- Sub-Saharan Africa is among the most vulnerable regions to climate change effects on agriculture.